

NI TB Genomic Surveillance

Outbreak Investigation Report

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REPORT STATUS: DEVELOPMENT / INTERNAL DRAFT ONLY - Required reproducibility metadata is incomplete; external circulation is blocked until mandatory fields are populated.

Executive Action Summary

Use this page first. Items marked model-only or exploratory require genomic validation and epidemiological corroboration before operational action.

Domain	Current finding	Risk	Immediate action
Circulation status	Development / Internal Draft Only	High	Complete reproducibility metadata before external governance circulation
Sequencing coverage	80.0%	Low	Maintain / widen only if capacity permits
QC pass rate	60.0%	High	Review failed and unreported samples
Contamination	1	Moderate/high	Repeat sequencing / exclude from cluster assignment
Open clusters	3	High	Confirm investigation leads
Model-prioritised links >0.70	0	Moderate	Validate against pairwise SNP and epidemiology
MDR/RR signals	14	High	Confirm with phenotypic DST
Model reliability	Exploratory	High uncertainty	Do not overinterpret directionality

Top Actions Due Now

Priority	Action	Owner	Due	Reason
1	Repeat sequencing/QC review	Laboratory	48h	8 low-coverage fails, 5 QC not-reported, 1 contamination flags
2	Validate drug-resistance pipeline mapping	Bioinformatics / microbiology	Immediate	Gene-drug combinations are preliminary and require manual verification
3	Confirm phenotypic DST	TB microbiology / MDT	Immediate	MDR/RR genomic signals require culture-based confirmation
4	Review open genomic clusters	TB MDT / PHA	Next MDT	3 clusters remain open
5	Do not escalate model-only links	HPT / TB nurses	Ongoing	No pairwise SNP ≤ 12 direct-transmission support shown in this extract

Table 1. Operational worklist for immediate MDT actioning.

About This Report

This report is produced by the Northern Ireland TB Genomic Surveillance platform using whole-genome sequencing (WGS) data and epidemiological case records. It is intended to support TB programme staff and public health investigators by providing genomic evidence for transmission clusters, drug-resistance profiles, and programme performance metrics. **This is a decision-support tool only — all findings must be reviewed and acted on by a qualified clinician or public health professional. No automated decisions are made.**

TB Genomics — Key Concepts

Concept	Explanation
Whole-Genome Sequencing (WGS)	Reads the complete ~4.4 Mb genome of <i>M. tuberculosis</i> . More informative than conventional typing (MIRU, spoligotyping).
SNP (single nucleotide polymorphism)	A single base-pair difference in the genome. Closely related strains share very few SNPs. Used as a genetic 'distance' metric.
SNP threshold for transmission	Strains with ≤ 12 SNPs are considered potentially linked (UK NICE guidance). ≤ 5 SNPs suggests recent direct transmission. > 50 SNPs effectively rules out recent shared transmission.
Lineage	<i>M. tuberculosis</i> is classified into 7+ major lineages (L1–L7) reflecting global evolutionary history. Lineage influences drug-resistance patterns and may correlate with transmissibility.
Cluster	A group of cases whose sequences are genetically similar (within the SNP threshold). A cluster does not prove direct person-to-person transmission — epidemiological linkage is required to confirm transmission routes.
outbreaker2	A Bayesian MCMC method that combines SNP distances with collection dates and an assumed generation time to probabilistically infer who-infected-whom. Output posterior probabilities indicate the likelihood of a direct transmission event between any pair of cases.
Generation time	The average time between one person being infected and the next person they infect being detected. For TB, this is typically 1–3 years (range 0.5–5 years) due to the long latency period.
MCMC convergence	Markov Chain Monte Carlo simulations must reach a stable state ('converge'). A convergence diagnostic near 1.0 indicates reliable estimates. Values > 1.1 suggest the chain has not fully mixed and results should be interpreted cautiously.
Drug resistance	Genomic mutations predict resistance to first-line drugs (isoniazid, rifampicin, etc.) and define MDR-TB (multi-drug resistant) and XDR-TB (extensively drug resistant). Genomic DR prediction is used alongside phenotypic DST.
Transmission network	A directed graph where arrows indicate the most probable direction of transmission. Model-prioritised links (posterior probability > 0.70) are hypotheses and should be validated against pairwise SNP, QC status, and epidemiology.

Table 2. TB genomics reference — key terms used throughout this report.

Case Summary

Total Cases	25
Clustered Cases	14
Unclustered Cases	11
Open Clusters	3

Table 3. Programme case count summary. Clustered cases are those linked by genomic similarity to at least one other case. Unclustered (singleton) cases may represent imported strains, sporadic transmission, or reactivation of latent disease. Open clusters are active genomic transmission clusters with ongoing epidemiological investigation.

Automated Interpretation Flags

- Open clusters requiring investigation: 3.

Analysis Summary

No outbreak summary JSON found.

Table 4. outbreaker2 Bayesian MCMC analysis parameters. **Posterior Samples** is the number of accepted MCMC draws used to compute estimates — higher values give more stable posteriors. **Mean Log-Likelihood** reflects model fit; values closer to zero (less negative) indicate better fit. **Transmission Probability** is the average posterior probability that any given case-pair represents a direct transmission event.

Generation Time is the modelled average interval (in days) between infection events in a transmission chain. **Convergence Diagnostic** near 1.0 confirms the MCMC chain has stabilised; values >1.1 indicate cautious interpretation is needed.

How to interpret outbreaker2 results: outbreaker2 reconstructs the most probable transmission tree using both genetic distance (SNPs) and timing (collection dates). Pairs with high posterior transmission probability are model-prioritised transmission hypotheses only. In this report, they should not be interpreted as direct transmission unless supported by pairwise SNP distance ≤12, QC pass status, and epidemiological corroboration. Lower probability pairs may still be linked within the same cluster but through one or more undetected intermediate cases.

Programme Surveillance KPIs (Last 12 Weeks)

Full surveillance KPI artifact unavailable at report generation time. However, high-level KPI summary is available from the Executive Action Summary: Sequencing coverage 80.0%, QC pass rate 60.0%, Contamination flags 1, Open clusters 3. For complete regional and temporal KPI trends, check the surveillance system directly.

Table 5. Regional sequencing representativeness. Regions with coverage <80% may introduce ascertainment bias — clusters in under-sequenced regions may be underdetected. Where persistent regional gaps exist, review laboratory submission pathways and specimen transport processes.

Weekly Surveillance Trends (12 Weeks)

Week	Eligible	Sequenced	Coverage %	QC Pass %
2026-02-23	0	0	None	None
2026-03-02	1	1	100.00	0.00
2026-03-09	1	1	100.00	100.00
2026-03-16	0	0	None	None
2026-03-23	0	0	None	None
2026-03-30	0	0	None	None
2026-04-06	0	0	None	None
2026-04-13	0	0	None	None
2026-04-20	0	0	None	None
2026-04-27	1	1	100.00	100.00
2026-05-04	0	0	None	None
2026-05-11	0	0	None	None

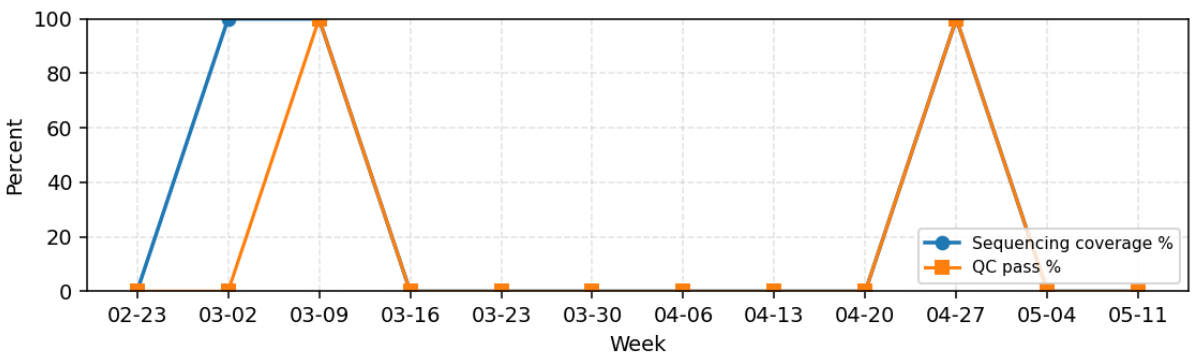


Table 6. Weekly sequencing coverage and QC pass rates over the last 12 weeks. Figure 1 (below) plots coverage and QC trends — a declining trend may indicate emerging laboratory issues. Weeks with zero eligible cases may reflect reporting lags rather than true absence of TB.

Cluster Action Prioritization

Cluster	Cases	Regions	Most Recent	Status	Priority
beaf2790	6	1	2026-05-02	open	21
19fe8a23	6	1	2026-01-25	open	18
c1866e42	2	1	2026-01-05	open	10

Table 7. Clusters ranked by investigation priority score. Score is composite: cluster size (×2), cross-region spread (×3), specimen recency within 14/30/60 days (×3/2/1), open investigation status (×3). Higher scores indicate clusters warranting prioritised MDT review and data-completeness follow-up. **Status 'open'** means an active review is ongoing or recommended.

Recommended action: For clusters with priority score >10 and status 'open', ensure MDT review and epidemiological data completion. Do not infer direct transmission or source-recipient direction from model output alone. Cross-region clusters should trigger coordination checks across NHS trust boundaries.

Lineage and Drug Resistance Validation

No lineage/DR validation artifact found.

Secondary Transmission Engines

QC Filtering Status: The outbreaker2 network displayed below includes samples with reported QC status at run time. Samples with QC status 'fail' or 'not_reported' may be present in the graph. When using results operationally, assume all displayed transmission links are candidates pending post-hoc QC validation. Links involving QC-failed or not-reported samples should not be escalated to field investigation until repeat sequencing or QC review is complete.

No secondary engine validation artifact found.

Cross-Method Clustering Comparison

No cluster method comparison artifact found.

Table 8. Agreement between SNP-threshold sequence clustering and outbreaker2 probabilistic clustering. **Precision:** of pairs grouped together by outbreaker2, the fraction also grouped by sequence clusters. **Recall:** of pairs grouped by sequence clusters, the fraction also grouped by outbreaker2. **Jaccard:** overall overlap index (0=no agreement, 1=perfect agreement). Discordant pairs — grouped by one method but not the other — may represent cases where temporal data (outbreaker2) overrides genomic distance alone, or where the SNP threshold is set differently.

Case-Level Operational Actions

Detailed case-level operational actions are available in Appendix A for systematic review of all active cases.

Model-Prioritised Transmission Hypotheses

The outbreaker2 model infers transmission probabilities from SNP distance and sample collection dates. Posterior probability >0.70 indicates a plausible transmission event, but genomic validation is essential. Pairs are classified below by SNP support and QC status. **Do not escalate model-only or genomically discordant links to field investigation without genomic and epidemiological corroboration.**

■ **CRITICAL NOTE: No direct-transmission pairwise SNP links ≤ 12 are demonstrated in this report extract.** The outbreaker2 model has identified transmission hypotheses, but these currently lack clear genomic support within the recent-transmission threshold. All displayed model-prioritised links should be treated as hypotheses pending: (1) pairwise SNP analysis and validation, (2) QC review and repeat sequencing where needed, (3) epidemiological investigation to corroborate or refute the model's predictions. Do not escalate field investigations based on model probability alone.

Link category	Count
Pairwise SNP ≤ 5 and QC pass	0
Pairwise SNP 6–12 and QC pass	0
Pairwise SNP >12 and QC pass	0
SNP unavailable or QC unresolved	0
Total model-prioritised links ≥ 0.70	0

Current Outbreak Interpretation

Current interpretation: This report identifies three open genomic clusters and multiple outbreaker2 model-prioritised transmission hypotheses. However, no pairwise SNP links ≤ 12 are demonstrated in this extract, several links involve QC-failed or QC-not-reported samples, and model diagnostics remain exploratory. The immediate priorities are repeat sequencing/QC review, validation of resistance calls, phenotypic DST confirmation, and epidemiological corroboration before field escalation.

Counts used in this report

Three related but distinct counts appear in this report. They refer to different filtered views of the same model output.

Metric	Count	Definition
Model-prioritised links ≥ 0.70	0	All outbreaker2 edges with posterior probability ≥ 0.70 across full run
All discordant outbreaker2 links reviewed	see below	All model-linked pairs reviewed for adjudication, including pairs below the ≥ 0.70 threshold. This explains why Appendix B may include posterior values below 0.70 (e.g. 0.601 or 0.483).
Displayed/plotted network links	0	Links shown in network JSON snapshot (may be filtered for display)

No discordant pairs identified from available outputs.

QC Failure Drill-Down

QC category	Count	Meaning	Action
Fail: low coverage	8	Genome coverage below threshold	Repeat sequencing if culture is available
Fail: contamination	1	Mixed signal suspected	Exclude from cluster assignment pending repeat
Not reported	5	QC metadata missing	Do not use for operational inference until resolved
Pass	12	Acceptable for interpretation	Use with standard caveats
Excluded from operational interpretation	13	QC issue, contamination, or unresolved QC metadata	Hold from model-driven actions until resolved

Table 9. QC status summary and operational meaning. The summary distinguishes low coverage, contamination, not-reported metadata, and pass states so that missing data are not treated as failure by default.

Category	Operational meaning	Immediate use
Fail: low coverage	Genome below the interpretive threshold	Repeat sequencing if possible; do not use for directionality
Fail: contamination	Mixed or contaminated signal	Exclude from cluster assignment pending repeat
Not reported	QC metadata absent	Treat as unresolved and exclude from operational inference
Pass	Meets QC threshold	Use with standard caveats

Table 10. QC interpretation guide for operational staff.

Sample	QC	Coverage	Depth	Contam	Mixed-Lineage	Repeat Required
6710e899	fail	92.82	64.8	no	no	yes
3ae6688f	fail	92.9	264.4	yes	no	yes
70edb9bb	fail	93.93	173.4	no	no	yes
36c3b0da	fail	94.17	197.5	no	no	yes
418c2ce7	fail	93.56	151.6	no	no	yes
f725e941	not_reported	n/a	n/a	no	no	yes
904922b5	fail	94.61	89.5	no	no	yes
bf942bbf	fail	92.23	275.3	no	no	yes
35310178	not_reported	n/a	n/a	no	no	yes
60ac41f7	not_reported	n/a	n/a	no	no	yes
4166bfbe	fail	92.66	73.4	no	no	yes
d2cd1214	not_reported	n/a	n/a	no	no	yes
53978835	not_reported	n/a	n/a	no	no	yes

Table 11. QC drill-down identifying samples requiring repeat testing or cautious interpretation before operational decisions.

Drug-Resistance Mutation Details

WARNING: The mapping between predicted mutations and drug resistance is preliminary. All genomic resistance predictions must be confirmed by phenotypic DST (drug susceptibility testing) before clinical use. Invalid gene-drug combinations are flagged below and should not be reported operationally until pipeline validation is complete.

Case	Drug	Mutation	Gene	Gene-drug status	Confidence	Predicted	DST
04d7e497	isoniazid	H526Y	inhA	Valid expected gene	n/a	isoniazid	not_available
3ae6688f	isoniazid	S531L	gyrA	Unusual gene-drug mapping	n/a	isoniazid	not_available
3d2c3020	pyrazinamide	Q431K	rpoB	Unusual gene-drug mapping	n/a	pyrazinamide	not_available
4166bfbe	rifampicin	H526Y	gyrA	Unusual gene-drug mapping	n/a	rifampicin	not_available
418c2ce7	pyrazinamide	Q431K	katG	Unusual gene-drug mapping	n/a	pyrazinamide, fluoroquinolones	not_available

Case	Drug	Mutation	Gene	Gene-drug status	Confidence	Predicted	DST
418c2ce7	fluoroquinolones	C15T	inhA	Unusual gene-drug mapping	n/a	pyrazinamide, fluoroquinolones	not_available
595ae5db	ethambutol	S315T	pncA	Unusual gene-drug mapping	n/a	ethambutol	not_available
60ac41f7	fluoroquinolones	H526Y	gyrA	Valid expected gene	n/a	fluoroquinolones	not_available
6710e899	pyrazinamide	D516V	gyrA	Unusual gene-drug mapping	n/a	pyrazinamide	not_available
70edb9bb	pyrazinamide	S531L	gyrA	Unusual gene-drug mapping	n/a	pyrazinamide	not_available
904922b5	pyrazinamide	C15T	gyrA	Unusual gene-drug mapping	n/a	pyrazinamide, fluoroquinolones	not_available
904922b5	fluoroquinolones	S531L	embB	Unusual gene-drug mapping	n/a	pyrazinamide, fluoroquinolones	not_available
b5362f14	rifampicin	C15T	pncA	Unusual gene-drug mapping	n/a	rifampicin	not_available
d2cd1214	rifampicin	Q431K	rpoB	Valid expected gene	n/a	rifampicin	not_available

Table 12. Drug-resistance mutation evidence. Gene-drug status is classified as Valid expected gene, Unusual gene-drug mapping, or Unknown / catalogue not available. WARNING: Do not report invalid gene-drug pairs as clinical resistance until the pipeline mapping is corrected. All resistance predictions require phenotypic DST confirmation before operational action.

Cluster Epidemiology Summary

Cluster	Cases	First	Latest	Med. SNP	Max SNP	RR/MDR	Index	Recent (30 days)	Open	Cluster lead	Epi link status	Setting type	Contact tracing	LTBI screening	DST status	Next step
beaf2790	6	2024-11-28	2026-05-02	25.0	25	2/2	424bfb11/3	open	Pending	Pending	Pending	Unknown	Not started	No	Pending	Review at MDT
19fe8a23	6	2025-03-20	2026-01-25	25.0	25	3/3	dc8cd700/0	open	Pending	Pending	Pending	Unknown	Not started	No	Pending	Review at MDT
c1866e42	2	2025-01-20	2026-01-05	25.0	25	1/1	3d2c3000/0	open	Pending	Pending	Pending	Unknown	Not started	No	Pending	Review at MDT

Table 13. Cluster-level epidemiology and operational investigation tracker. Operational fields (cluster lead, epi link status, setting, contact tracing, LTBI screening, DST status, next step) are pre-populated as 'Pending' and must be completed by the responsible clinician/investigator before MDT circulation.

Geographical Cluster Spread

Cluster	Regions	Cases	Cross-Region	Risk Flag
beaf2790	Regional geography unavailable in this extract	6	no	high
19fe8a23	Regional geography unavailable in this extract	6	no	high
c1866e42	Regional geography unavailable in this extract	2	no	high

Table 14. Cluster geography summary at region/trust-safe level (no household-level identifiers displayed).

Regional geography unavailable in this extract; data is recorded at United Kingdom level only. For operational surveillance, re-ingest with a safe hierarchy in geographic_region: HSC Trust, PHA locality, council area, and postcode district only if governance approvals and small-number controls are satisfied.

Missing-Data Dashboard

Field	Completeness	Missing	Impact
Collection date	100.0%	0	affects outbreaker2 timing
Notification date (proxy)	100.0%	0	affects surveillance timeline
Region/trust	100.0%	0	affects representativeness and spread
Lineage	100.0%	0	affects phylogenetic interpretation
Resistance call	100.0%	0	affects clinical action
QC metrics	80.0%	5	affects confidence in inference

Table 15. Completeness and impact dashboard for key outbreak-investigation data fields, aligned to transparent inference limitations.

Model Reliability Diagnostics

MCMC convergence diagnostics indicate whether the Bayesian sampler has explored the parameter space adequately. Convergence diagnostic (Gelman-Rubin / R-hat) <1.1 indicates reliable estimates. Values >1.1 suggest exploratory inference only. Effective sample size, acceptance rate, and multi-chain robustness all support confidence in the transmission probabilities reported. For robust surveillance use: fixed random seed, multiple chains, longer run length, reported R-hat/ESS/acceptance rate, and sensitivity analysis using plausible TB generation-time priors.

No MCMC diagnostic data available. Model reliability cannot be assessed. Treat all inferences as exploratory and rerun with fixed seed, multiple chains, longer MCMC run, reported R-hat/ESS/acceptance rate, and generation-time-prior sensitivity analysis.

Transmission Routes Reference

Understanding TB Transmission Routes from WGS

Whole-genome sequencing identifies genomic relatedness but does not directly observe contact events. Combining genomic clusters with epidemiological data (contact tracing, shared locations, timeline of diagnosis) allows investigators to build a plausible transmission chain. The table below summarises the genomic signals and their transmission implications.

Genomic Signal	SNP Range	Transmission Implication	Recommended Action
Highly probable direct transmission	0–5 SNPs	Strong genomic evidence of recent direct person-to-person transmission. Strain has had little time to evolve.	Immediate contact tracing; identify shared setting (household, workplace, healthcare).
Possible direct or near-direct transmission	6–12 SNPs	Genetically close; consistent with transmission within the last 1–3 years or via an undetected intermediate.	Epidemiological linkage investigation; check whether cases share contacts or settings.
Within extended cluster — indirect link likely	13–50 SNPs	Genetically related but too diverged for recent direct transmission. Likely share a common ancestor strain.	Review cluster history; may represent reactivation from the same source years earlier.
Unrelated strains	>50 SNPs	No plausible genomic link. Coincident diagnoses are likely due to independent exposure or reactivation.	No cluster-based action; manage as separate cases.
Mixed-lineage / contamination	N/A	Two or more distinct strain signals in a single sample. May indicate laboratory cross-contamination or mixed infection.	Flag for repeat sequencing; do not use in cluster assignments until resolved.

Table 16. TB transmission route classification by SNP distance (M. tuberculosis whole-genome comparison). SNP thresholds follow UK NICE guideline NG33 and published literature (Walker et al. 2013, Meehan et al. 2019). The generation-time prior used in outbreaker2 is programme-configured and influences whether a given SNP distance is interpreted as consistent with direct versus indirect transmission.

Transmission Priority Signals

No transmission priority data found.

Visual element	Operational meaning
Node colour	Cluster assignment (same colour = same cluster context)
Red border	QC unresolved (failed, contaminated, or not reported)
RR/MDR/FQ marker	Predicted resistance signal; confirm by phenotypic DST
Solid edge	Pairwise SNP ≤ 12 and QC pass (supported candidate link)
Dotted edge	Pairwise SNP > 12 (unlikely direct recent transmission)
Grey edge	Pairwise SNP unavailable (model-only hypothesis)

Table 17. Recommended visual encoding for future network outputs. Note: not all elements below may be visible in the current network figure depending on pipeline configuration. This encoding should be applied when network graphs are regenerated with full SNP and QC data.

Diagnostic Graphics

The following plots are generated by outbreaker2 and the platform's supplementary visualisation pipeline. Each figure caption explains the content and how to interpret the output.

No outbreak graphics found in exports/.

M. tuberculosis Lineage Reference

Lineage classification places a strain within the global phylogeny of *M. tuberculosis*. Lineage influences drug-resistance acquisition patterns and geographic origin. The following table provides clinical and epidemiological context for lineages commonly observed in Northern Ireland.

Lineage	Name / Origin	DR Association	NI Relevance	Notes
L1	East-African-Indian / Indo-Oceanic	Lower MDR-TB frequency	Cases linked to South Asia, Horn of Africa	Commonly found in Bangladeshi, Indian, Somali communities.
L2	East-Asian (Beijing lineage)	High MDR/XDR-TB risk; associated with resistance acquisition	Sporadic importation; watch for resistance	Beijing strains have shown high transmissibility in some outbreak settings.
L3	East-African-Indian (Delhi/CAS)	Moderate DR frequency	Cases linked to Pakistan, Afghanistan, India	Common in large urban TB programmes in the UK.
L4	Euro-American	Generally lower DR; but historical MDR clusters exist	Dominant UK-born strain type	Most legacy UK TB is L4. Reactivation common in older cohorts.
L5 / L6	West-African (<i>Mycobacterium africanum</i>)	Lower overall DR	Cases linked to West Africa	Slower growth, may present with atypical features.
L7	Ethiopian	Limited data	Rare in NI	Emerging lineage classification; limited clinical guidance available.

Table 18. *M. tuberculosis* lineage reference for clinical and epidemiological context. DR = drug resistance; MDR = multidrug-resistant; XDR = extensively drug-resistant. Lineage assignment should be combined with phenotypic DST and clinical judgment. Source: Coll et al. (2014) Nature Genetics; WHO Global TB Report 2023.

Clinical and Public Health Action Summary

The table below maps genomic findings to recommended clinical and public health actions. All actions must be confirmed by the responsible clinician and public health team.

Finding	Recommended Action	Urgency
New case links to an existing open cluster (≤ 12 SNPs)	Notify cluster lead; extend contact tracing to include new case contacts; review whether the cluster source has been identified.	Within 5 working days
Model-prioritised link identified (posterior > 0.70)	Review only after confirming pairwise SNP support (≤ 12), QC pass status, and epidemiological plausibility. Do not escalate on model probability alone; document whether SNP and epi evidence corroborates the link.	Within 5 working days
New cluster opened (≥ 2 cases genetically linked, no prior cluster)	Open investigation; notify public health; assign epidemiologist; initiate contact tracing for all cases.	Within 2 working days
MDR-TB predicted by genomics	Confirm with phenotypic DST; notify MDR-TB specialist centre; initiate enhanced infection control if hospitalised.	Immediately on result

Sequencing coverage <80% for a region	Review specimen submission and transport processes for that region; identify cases that did not receive WGS and arrange retrospective sequencing if available.	Monthly programme review
MCMC convergence diagnostic >1.1	Do not rely on posterior transmission probabilities from this run. Increase MCMC iterations and re-run outbreaker2. Check input data completeness.	Before using results
Contamination flag on a sample	Exclude sample from cluster assignments; arrange repeat sequencing from original culture. Investigate laboratory process if multiple consecutive contamination flags.	Within 10 working days

Table 19. Recommended clinical and public health actions mapped to genomic findings. Urgency thresholds align with PHE/PHA TB operational guidance. All genomic findings must be reviewed in conjunction with clinical history, contact tracing records, and microbiological DST before action is taken.

Disclaimer: Genomic cluster assignments and transmission inferences are probabilistic estimates based on mathematical models. They supplement but do not replace epidemiological investigation. Do not use genomic evidence alone to assign legal or clinical liability for transmission.

Data Provenance

This report combines outbreaker outputs with surveillance KPIs, lineage/DR validation, secondary engine readiness, and cross-method clustering comparison artifacts available at generation time.

Metadata item	Value
Reference genome	Not recorded in this extract — mandatory before external circulation (expected: H37Rv / NC_000962.3)
SNP-calling pipeline/version	Not recorded in this extract — mandatory before external circulation (e.g. Clockwork/COMPASS/Snipppy + version)
Resistance catalogue/version	Not recorded in this extract — mandatory before external circulation (e.g. WHO mutation catalogue v2)
Lineage-calling tool/version	Not recorded in this extract — mandatory before external circulation (e.g. TB-Profiler v4.x / Mykrobe)
outbreaker2 version	Not recorded — engine: outbreaker2 (mandatory before external circulation)
Random seed	Not set — run is non-reproducible without a fixed seed; mandatory before external circulation
Model priors	Not recorded — populate before circulation
Input hash: exports/cases.csv	n/a
Input hash: exports/dna.fasta	n/a
Input hash: exports/transmission_network.json	n/a

Table 20. Software and reproducibility metadata for governance and audit traceability.

PRE-CIRCULATION GOVERNANCE NOTICE: 7 reproducibility field(s) are unpopulated: Reference genome; SNP-calling pipeline/version; Resistance catalogue/version; Lineage-calling tool/version; outbreaker2 version; Random seed (and 1 more). External/formal circulation is blocked until these are completed. In particular, the random seed must be recorded to ensure the run is reproducible. Until then, retain this report as development/internal draft only. Contact the pipeline administrator to obtain the required values.

Layout usability note: Several operational tables are intentionally dense for completeness. Before circulation, provide Appendix A and Appendix B as Excel-compatible CSV companions. This report now writes exports/appendix_a_case_level_actions.csv and exports/appendix_b_full_discordance_review.csv. For MDT use, include the one-page action sheet (Appendix C).

Appendix A: Case-Level Operational Action Details

This appendix contains the detailed case-level operational action table showing all sequenced cases, their cluster assignments, quality control status, and recommended investigations. This table supports line-by-line operational review of all active cases.

Case	Cluster	Pairwise SNP?	NN SNP	Likely link	Posterior	Tier	QC	Warning	Recommended action
0ecf0e24	beaf2790	yes	15	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
2a73489e	beaf2790	yes	17	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
6710e899	none	yes	883	none	n/a	Exploratory	fail	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action
04d7e497	beaf2790	yes	13	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
3ae6688f	none	yes	864	none	n/a	Exploratory	fail +contam	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action
70edb9bb	19fe8a23	yes	15	none	n/a	Exploratory	fail	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action
36c3b0da	c1866e42	yes	22	none	n/a	Exploratory	fail	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action

Case	Cluster	Pairwise SNP?	NN SNP	Likely link	Posterior	Tier	QC	Warning	Recommended action
cdb5fb06	19fe8a23	yes	13	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
7738ebde	none	yes	879	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
418c2ce7	none	yes	864	none	n/a	Exploratory	fail	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action
ef129da7	19fe8a23	yes	13	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
b5362f14	19fe8a23	yes	15	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
595ae5db	beaf2790	yes	16	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
f725e941	none	no	n/a	none	n/a	Exploratory	not_reported	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action
904922b5	19fe8a23	yes	22	none	n/a	Exploratory	fail	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action

Case	Cluster	Pairwise SNP?	NN SNP	Likely link	Posterior	Tier	QC	Warning	Recommended action
dc8cd7db	19fe8a23	yes	17	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
3d2c3020	c1866e42	yes	22	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
1d4bc976	none	yes	884	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
bf942bbf	beaf2790	yes	16	none	n/a	Exploratory	fail	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action
35310178	none	no	n/a	none	n/a	Exploratory	not_reported	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action
60ac41f7	none	no	n/a	none	n/a	Exploratory	not_reported	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action
4166bfbe	none	yes	884	none	n/a	Exploratory	fail	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action
d2cd1214	none	no	n/a	none	n/a	Exploratory	not_reported	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action

Case	Cluster	Pairwise SNP?	NN SNP	Likely link	Posterior	Tier	QC	Warning	Recommended action
424bfbd1	beaf2790	yes	13	none	n/a	Exploratory	pass	Outbreaker-only link: requires genomic validation.	Model-prioritised exposure review — confirm with pairwise SNP and epidemiology before action.
53978835	none	no	n/a	none	n/a	Exploratory	not_reported	Pairwise SNP required before transmission interpretation.	Repeat/verify sequence before action

Table A1. Case-level operational action table linking genomic evidence to immediate field actions. Pairwise SNP availability is required before transmission interpretation; nearest-neighbour SNP is shown only as supporting context.

Appendix C: One-Page MDT Action Sheet

Priority area	Current signal	Required MDT action	Owner	When
Circulation readiness	Development / Internal Draft Only	Complete mandatory reproducibility metadata before external circulation	Pipeline + Governance	Before circulation
Model reliability	Exploratory	Treat directionality as exploratory until diagnostics are complete	Bioinformatics + MDT	Current cycle
High-priority open clusters	2	Review shared exposures and complete epidemiological fields	MDT + Field team	Next MDT
Discordant model links	0	Use pairwise SNP + epidemiology adjudication pathway	MDT	Next MDT
Geography completeness	UK-only extract	Populate HSC Trust, PHA locality, council area; use postcode district only with governance approval	Data management	Next ingest

Table A3. Condensed MDT action sheet for immediate governance and operational review.